

Professional Practices- week4

Cyber Crime, Intellectual property



- ❑ CYBER CRIME.**
- ❑ FORMS OF CYBER CRIME.**
- ❑ HACKING VS CRACKING.**
- ❑ INTELLECTUAL PROPERTY.**
- ❑ TYPES OF INTELLECTUAL PROPERTY.**

Cyber Crime



We can define cyber crime as :-

Any kind of crime or criminal act in which a computer /Mobile /cellular phone/tablet/landline phone is used as the principal tool

Forms of Cyber Crime



- ❑ Fraud.
- ❑ Embezzlement.
- ❑ Software/Information Sabotage.
- ❑ Information (Data) Theft.
- ❑ Hacking and Cracking.
- ❑ Digital Forgery.

Fraud



Wrongful or criminal act intended to result in financial or personal gain.

Example:-

- A bank staff or bank accountant gets the money for salary of staff in his account from bank and runs away.
- A cashier takes cash from super market and don't put amount in POS terminal etc..

Embezzlement



when someone with money-handling duties accepts a payment but doesn't enter the transaction in computer, and keeps the money himself.

Example:-

- A cashier takes cash from super market and don't put amount in POS terminal etc..

Fraud and Embezzlement up to Dec 2022



- ❑ Losses from credit card fraud about - \$2 ~ \$6 billion per year.
- ❑ ATM fraud accounts losses - about \$80 million a year.
- ❑ Telecommunications frauds - \$2 to \$11 billion each year.

Software/Information Sabotage



- ❑ An intentional attempt to damage (corrupt) a software or data.
- ❑ E.g. Infecting a website.
- ❑ Sending virus to hit MBR or EXE file.

Information (Data) Theft



- ❑ It is the illegal transfer or storage of personal, confidential, or financial information.
- ❑ This is done by disclosing IDs, keys or passwords OR unauthorized access to computers
- ❑ Infecting someone's used id or password.

Hacking



- ❑ It is an attempt to get information of a computer system or a computer network.
- ❑ It is the unauthorized access to a computer and its information or a computer network for some illegal purpose.
- ❑ The person who performs hacking is called a Hacker.
- ❑ Hacking is performed by using computer programs.
- ❑ Hackers are intelligent computer programmers using their minds negatively.

Cracking



- ❑ Cracking: Try to get into computer systems in order to steal, corrupt, or illegally view data.
- ❑ The person who performs cracking is called a Cracker.
- ❑ Crackers are not computer programmers, they use other techniques to break user ID Codes, Pin Codes and Passwords.

Hacking VS Cracking



- In Hacking, hackers access computer systems or networks but they may or may not harm.
- In Cracking, crackers access computer systems to steal, corrupt, or illegally view data.
- Hackers are computer programmers but Crackers are not.

Example of Hacking



- ❑ CAM4 Data Breach. Date: March 2020. ...
- ❑ Yahoo Data Breach (2017) Date: October 2017. ...
- ❑ Alibaba Data Breach. ...
- ❑ First American Financial Corporation Data Breach. ...
- ❑ Verifications.io Data Breach. ...
- ❑ LinkedIn Data Breach (2021) ...
- ❑ Facebook Data Breach (2019)

Digital Forgery (Digital Tampering)



- It is the process of manipulating documents or images for the purpose of financial, social or political gain.
- It is the criminal act that provides misleading information about a product or service.

Security Measures



- ❑ Electronically track and record the access to, and activities of, the various users of a computer system with the help of servers.
- ❑ Assign an individual ID and password to each user of the system.
- ❑ Password should not be simple, it should be combination of alphabetic, numeric and special characters.

Security Measures (continues)



- Each user should change password from time to time.
- Only authorized people should be allowed to enter in servers rooms and places where computers are used.
- Log files should be maintained of the people entering and going out of the sensitive areas having information like date, time-in, time-out, name of person, purpose etc.

Intellectual Property



- ❑ It is the creations of the human mind, whether creative or intellectual (Intelligent), such as art and designs, literary works, music, and the names, symbols, and images used in business

Intellectual Property Examples



Examples:

- ☐ Art: painting.
- ☐ Design: building design.
- ☐ Literary work: novel
- ☐ Music: new song
- ☐ Program, Flow charts, Algorithm, Book, Research Papers.
- ☐ Symbol: red light on signal means “STOP”.
- ☐ Image: scenery.

Types of Intellectual Property



Intellectual Property are of four types :-

- ❑ Copyright
- ❑ Trademark
- ❑ Patent
- ❑ Trade Secrets

Copyright



- ❑ It protect items and does not allow to copy or make exactly similar items.
- ❑ Applies to books, periodicals (journals, bulletins, magazines, letters etc.), songs, music, theatre productions (dramas, variety programs, talk shows etc.), movies etc.

Copyright Act



The basic legal instrument governing copyright law in Pakistan is the Copyright Ordinance, 1962 as amended by the Copyright (Amendment) Ordinance, 2000.

Trademark



- ❑ It is an exclusive symbol used for an item or company.
- ❑ E.g. trademark of McDonald's
- ❑ Trademark speaks of the product itself.
- ❑ It is illegal to copy a trademark, hence a person or company can file a case in the court against the copier.



Patent



- ❑ A government authority permits to produce a specific item, so that the producer can make, use, and sell for a specific time period.
- ❑ Only the party who has been given permission can do the business of that item, no other party is allowed to make, use or sell the item.
- ❑ Example 1:- “Jam-e-Shirin” is produced by Qarshi.
- ❑ Example 2:- “Rooh Afza” is produced by “Hamdard Pakistan”, as Pakistani government has given permission to produce it.
- ❑ Example 3:- “Student Biryani”, “Mazaidar Haleem” etc..

Trade Secret



- ❑ If a company can keep a trade secret, it will have exclusive rights to that product forever.
- ❑ Generally the trade secrets are the formulas which make a product exclusive than other products.
- ❑ Example: The longest running trade secret is the formula for Coca-Cola. The Coca-Cola Corporation never applied for a patent. They relied on their trademark secret. Although many companies have tried to produce the taste of Coca-Cola but failed to do so.



END